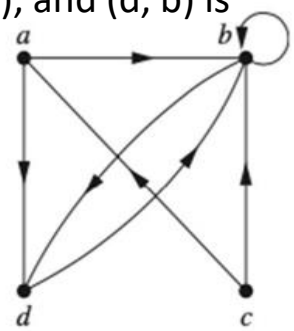
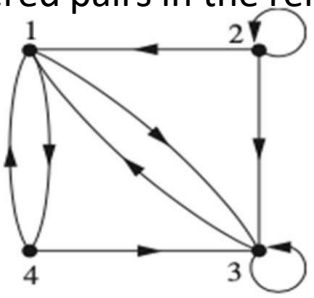
Representing Relations Using Digraphs

Definition: A directed graph consists of a set V of vertices (or nodes) together with a set E of edges (or arcs). If (a, b) is in the relation, then there is an arrow from a (initial vertex) to b (terminal vertex). An edge of the form (a, a) is represented using an arc from the vertex a back to itself. Such an edge is called a loop.

Example: The directed graph with vertices a, b, c, and d, and edges (a, b), (a, d), (b, b), (b, d), (c, a), (c, b), and (d, b) is



Example: What are the ordered pairs in the relation R represented by the directed graph?

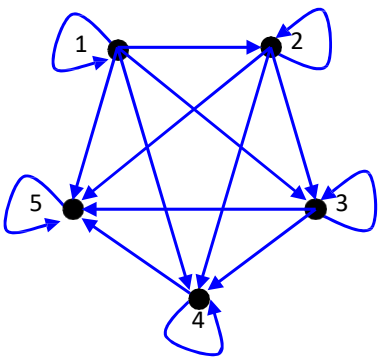


The ordered pairs (x, y) in the relation are

R={(1,3),(1,4),(2,1),(2,2),(2,3),(3,1),(3,3),(4,1),(4,3)}.

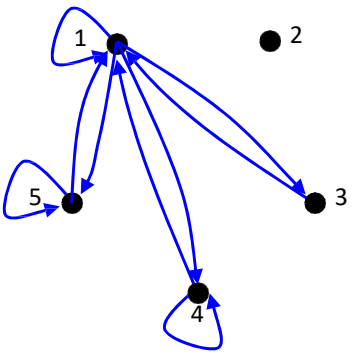
Determining which Properties a Relation has from its Digraph

Consider a relation, (a,b) ∈ R if a ≤ b. One which every element is related to itself.
Let A = { 1, 2, 3, 4, 5 }



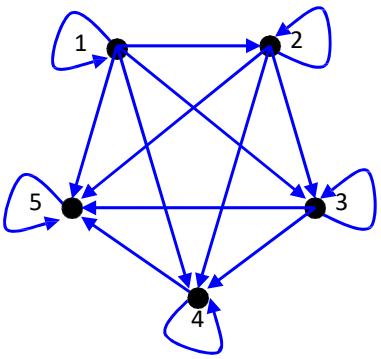
If every node has a loop, a relation is reflexive

Consider an symmetric relation R. One which if a is related to b then b is related to a for all (a,b). Let A = { 1, 2, 3, 4, 5 }



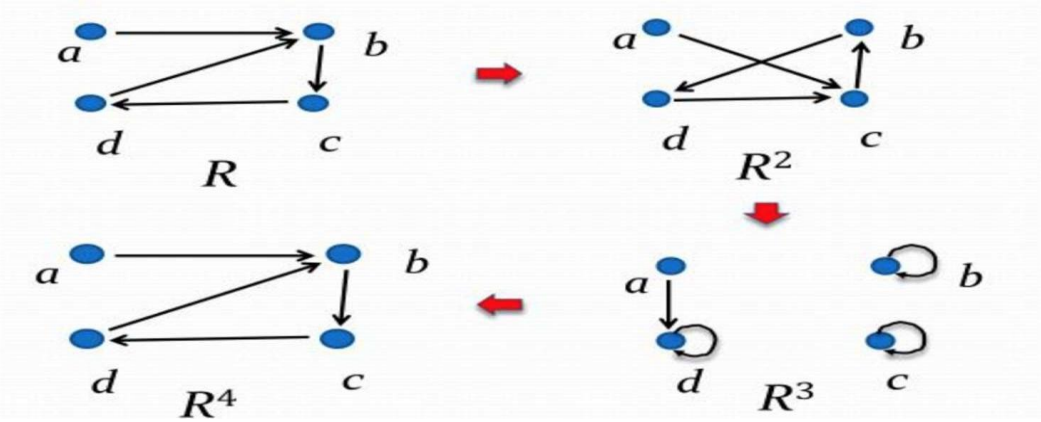
If, for every edge, there is an edge in the other direction, then the relation is symmetric . Loops are allowed, and do not need edges in the "other" direction

Consider a relation, (a,b) ∈ R if a ≤ b. One which every element is related to itself. Let A = { 1, 2, 3, 4, 5 }



If, for every edge, there is *not* an edge in the other direction, then the relation is antisymmetric . Loops are allowed in the digraph

Example of the Powers of a Relation



The pair (x,y) is in R^n if there is a path of length n from x to y in R^n (following the direction of the arrows).

Graph Theory

Definition: A graph G = (V, E) consists of a nonempty set V of vertices (or nodes) and a set E of edges (or arcs). Each edge is associated with either one or two vertices called endpoints. An edge is said to connect its endpoints. This is called undirected graph.

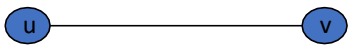
Remark: The set of vertices *V* of a graph *G* may be infinite. A graph with an infinite vertex set or an infinite number of edges is called an **infinite graph**, and in comparison, a graph with a finite vertex set and a finite edge set is called a **finite graph**. In this book we will usually consider only finite graphs

Graph Terminology

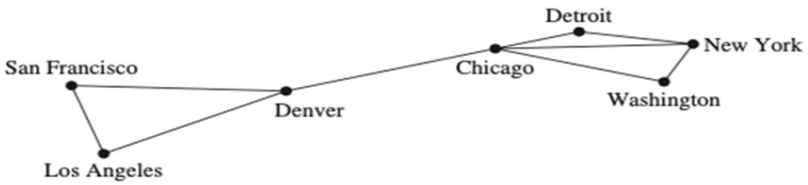
Directed: Ordered pair of vertices. Represented as (u, v) directed from vertex u to v.



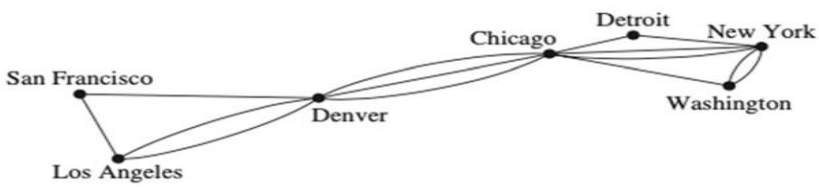
Undirected: Unordered pair of vertices. Represented as {u, v}. Disregards any sense of direction and treats both end vertices interchangeably.



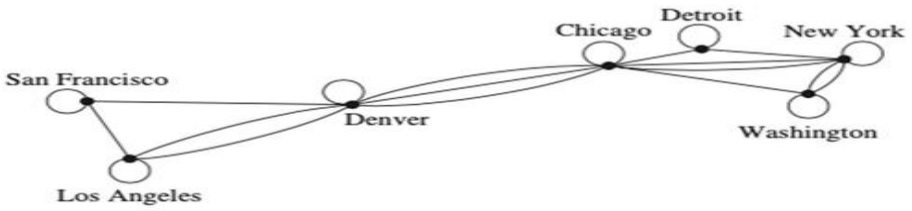
Simple (Undirected) Graph: consists of V, a nonempty set of vertices, and E, a set of unordered pairs of distinct elements of V called edges (undirected).



Multigraph: G(V,E), consists of set of vertices V, set of Edges E and a function f from E to {{u, v}| u, v V, u ≠ v}. The edges e1 and e2 are called multiple or parallel edges if f (e1) = f (e2).



Pseudograph: G(V,E), consists of set of vertices V, set of Edges E and a function F from E to {{u, v}| u, v ∈ V}. Loops allowed in such a graph. A **loop** is an edge whose endpoints are equal i.e., an edge joining a vertex to it self is called a loop. Represented as {u, u} = {u}



Simple (Directed) Graph: consists of V, a nonempty set of vertices, and E, a set of ordered pairs of distinct elements of V called edges (directed).

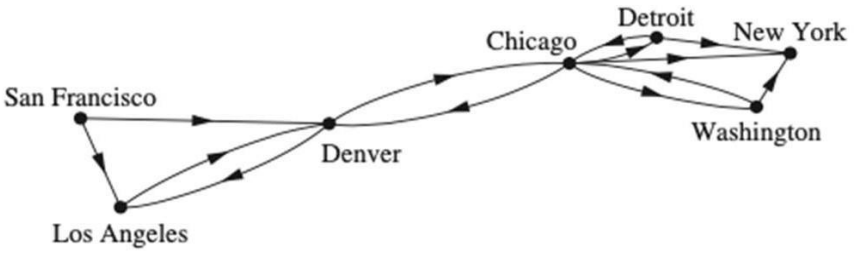


FIGURE 4 **A communications network with one-way communications links.**

Directed graphs that may have multiple directed edges from a vertex to a second (possibly the same) vertex are used to model such networks. We called such graphs **Directed Multigraph**.

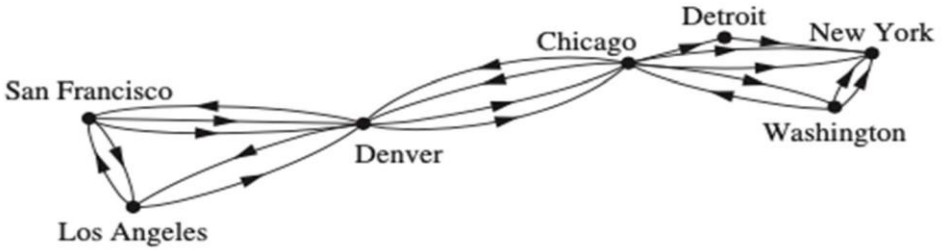
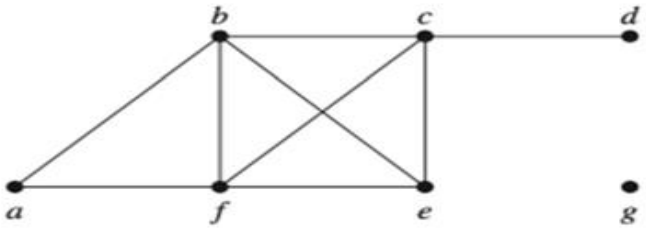


FIGURE 5 **A computer network with multiple one-way links.**

Graph Terminology - Undirected

- u and v are **adjacent** if {u, v} is an edge, e is called **incident** with u and v. u and v are called **endpoints** of {u, v}.
- The set of all **neighbours** of a vertex v of G = (V, E), denoted by N(v), is called the neighbourhood of v . If A is a subset of V, we denote by N(A) the set of all vertices in G that are adjacent to at least one vertex in A.
- **Degree of Vertex (deg (v)):** the number of edges incident on a vertex. A loop contributes twice to the degree (why?).
- **Pendant Vertex:** deg (v) =1
- **Isolated Vertex:** deg (v) = 0

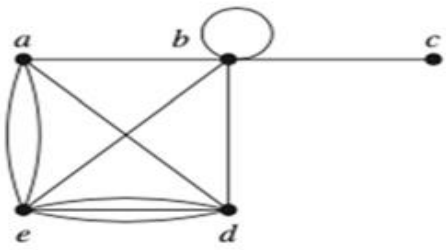
Example: What are the degrees and what are the neighborhoods of the vertices in the graphs G and H ?



G

deg(a) = 2,
deg(b) = deg(c) = deg(f) = 4, deg(d) = 1,
deg(e) = 3, and
deg(g) = 0.

The neighborhoods of these vertices are
N (a) = {b, f},
N (b) = {a, c, e, f},
N (c) = {b, d, e, f},
N (d) = {c}, N (e) = {b, c, f},
N (f) = {a, b, c, e}, and N (g) ={φ}.



H

deg(a) = 4,
deg(b) = deg(e) = 6, deg(c) = 1, and
deg(d) = 5.

The neighborhoods of these vertices are
N (a) = {b, d, e},
N (b) = {a, b, c, d, e},
N (c) = {b}, N (d) = {a, b, e}, and N
(e) = {a, b, d}

Degree of Vertices

Theorem: If G=(V,E) is an undirected graph with m edges, then

$$2m = \sum_{v \in V} \deg(v)$$

Theorem 1 shows that the sum of the degrees of the vertices of an undirected graph is even.

Proof: Each edge contributes twice to the degree count of all vertices.
Hence, both the left-hand and right-hand sides of this equation equal twice the number of edges.

Think about the graph where vertices represent the people at a party and an edge connects two people who have shaken hands.

Example: How many edges are there in a graph with 10 vertices of degree six?

Solution: Because the sum of the degrees of the vertices is 6 × 10 = 60, the handshaking theorem tells us that 2m = 60.So the number of edges m = 30.

Example: If a graph has 5 vertices, can each vertex have degree 3?

Solution: This is not possible by the handshaking theorem, because the sum of the degrees of the vertices 3 × 5 = 15 is odd.

Theorem: An undirected graph has an even number of vertices of odd degree.

Proof: Let V1 and V2 be the set of vertices of even degree and the set of vertices of odd degree, respectively, in an undirected graph G = (V, E) with m edges. Then

$$2m = \sum_{v \in V} \deg(v) = \sum_{v \in V_1} \deg(v) + \sum_{v \in V_2} \deg(v).$$

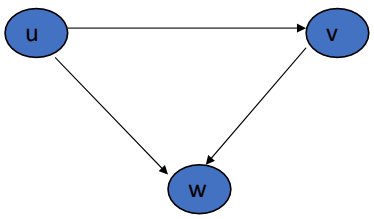
Because deg(v) is even for v $\in$ V1, the first term in the right-hand side of the last equality is even. Because 2m is even then the right-hand side is also even. Hence, the second term is also even:

$$\sum_{v \in V_2} \deg(v) = even$$

Graph Terminology - Directed

- For the edge (u, v), u is **adjacent to** v OR v is **adjacent from** u, u – **Initial vertex**, v – **Terminal vertex**
- **In-degree (deg^- (u)):** number of edges for which u is terminal vertex
- **Out-degree (deg^+ (u)):** number of edges for which u is initial vertex
- *Note: A loop contributes 1 to both in-degree and out-degree (why?)*

Example: In the graph G we have V = {u, v, w} , E = { (u, w), (v, w), (u, v) },
deg- (u) = 0, deg+ (u) = 2,
deg- (v) = 1, deg+ (v) = 1,
deg- (w) = 2, deg+ (w) = 0



Degree of Vertices

Theorem: If G=(V,E) is a directed graph, then

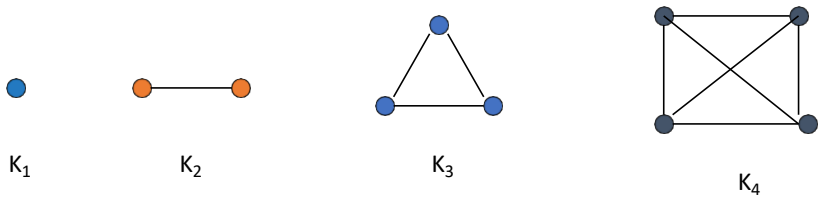
$$|E| = \sum_{v \in V} deg^-(v) = \sum_{v \in V} deg^+(v).$$

Proof: Because each edge has an initial vertex and a terminal vertex, the sum of the in-degrees and the sum of the out-degrees of all vertices in a graph with directed edges are the same. Both of these sums are the number of edges in the graph.

Simple graphs – special cases

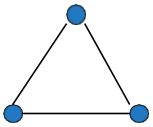
- **Complete graph:** K_n, is the simple graph that contains exactly one edge between each pair of distinct vertices.

  Representation Example: K_1, K_2, K_3, K_4

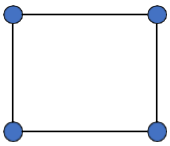


- **Cycle:** C_n, $n \geq 3$ consists of n vertices $v_1, v_2, v_3 \ldots v_n$ and edges $\{v_1, v_2\}$, $\{v_2, v_3\}$, $\{v_3, v_4\} \ldots \{v_{n-1}, v_n\}$, $\{v_n, v_1\}$

  Representation Example: C_3, C_4



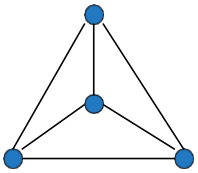
C_3



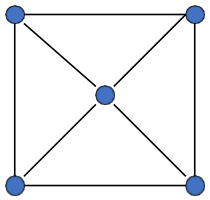
C_4

- **Wheels:** W_n, obtained by adding additional vertex to Cn and connecting all vertices to this new vertex by new edges.

  Representation Example: W_3, W_4



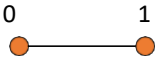
W_3



W_4

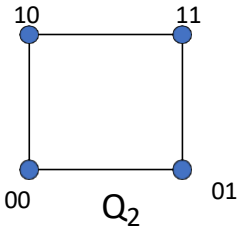
- **N-cubes:** Q_n, vertices represented by 2^n bit strings of length n. Two vertices are adjacent if and only if the bit strings that they represent differ by exactly one bit positions

  Representation Example: Q_1, Q_2



Q_1

n = 2



Q_2

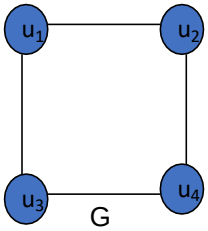
n = 3

Isomorphism of Graphs

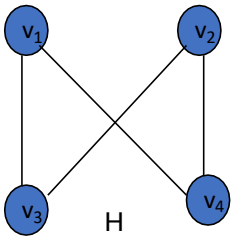
Definition: G1 = (V1, E2) and G2 = (V2, E2) are isomorphic if there is a one-to-one and onto function f from V1 to V2 with the property that a and b are adjacent in G1 if and only if f (a) and f (b) are adjacent in G2, for all a and b in V1. Function f is called isomorphism.

Example: G1 = (V1, E1) , G2 = (V2, E2)

$f(u_1) = v_1$, $f(u_2) = v_4$, $f(u_3) = v_3$, $f(u_4) = v_2$, is a one-to-one correspondence between V and W.



G

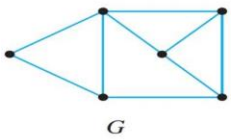


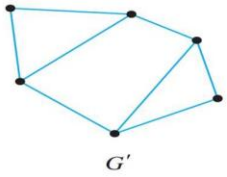
H

To see that this correspondence preserves adjacency, note that adjacent vertices in G are {u1 and u2}, {u1 and u3}, {u2 and u4}, and {u3 and u4}, and each of the pairs {f(u1) = v1 and f(u2) = v4}, {f(u1) = v1 and f(u3) = v3}, {f (u2) = v4 and f(u4) = v2}, and {f(u3) = v3 and f(u4) = v2} consists of two adjacent vertices in H.

- Unfortunately, there is no more efficient general method known for checking whether two graphs are isomorphic.
- However, there are some simple tests that can be used to show that certain pairs of graphs are not isomorphic.
- For instance, if two graphs are isomorphic, then they have the same number of vertices (because of the one-to-one correspondence on vertex sets of the two graphs).
- It follows that if you are given two graphs, one with 16 vertices and the other with 17, you can immediately conclude that the two are not isomorphic.

- **Example:** Show that the following pairs of graphs are not isomorphic by finding an isomorphic invariant that they do not share.





- G has nine edges;
- G′ has only eight.
- H has a vertex of degree 4;
- H′ does not.

